

Computational Wind Engineering

Bridge Aerodynamics

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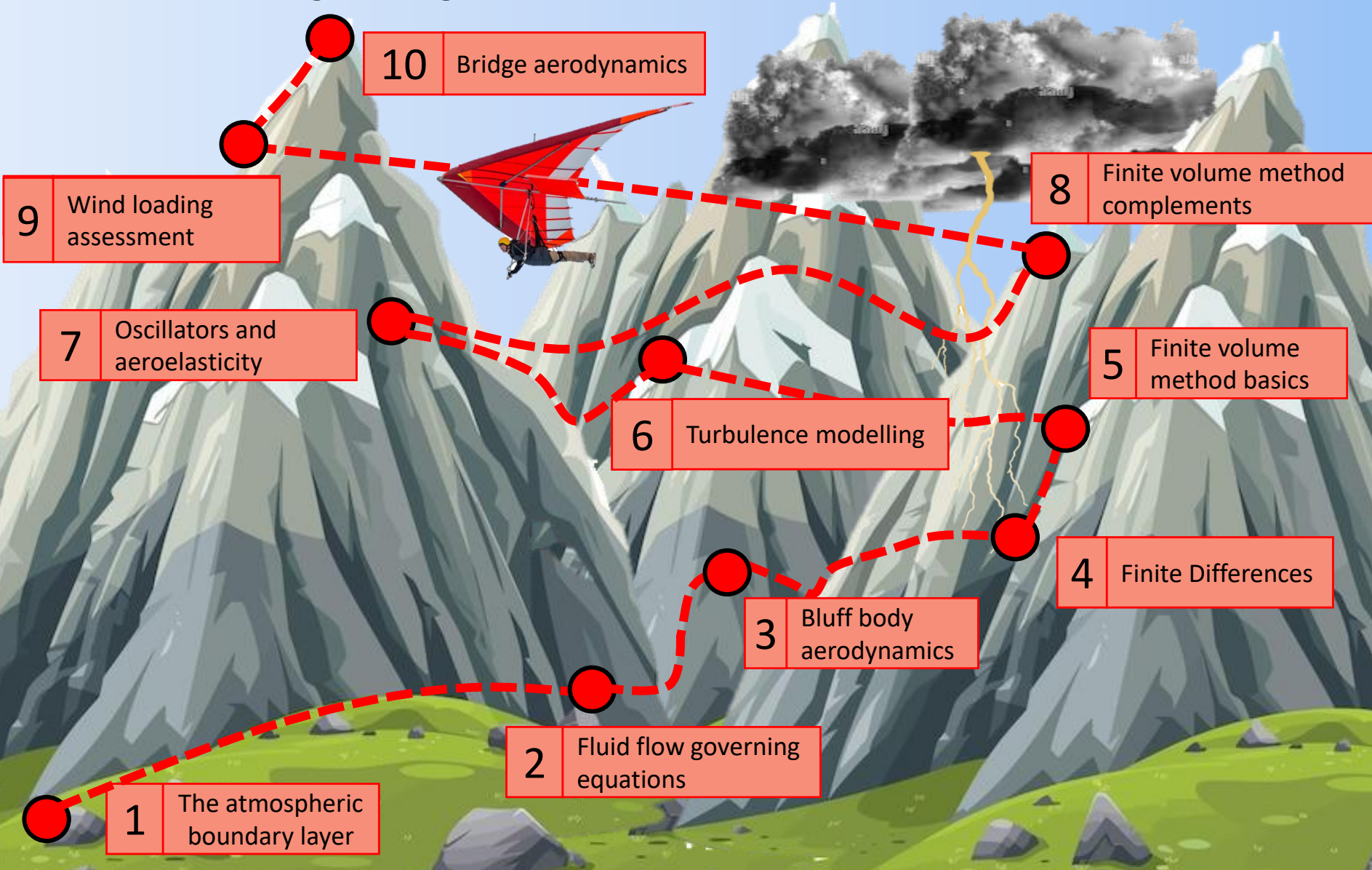
- Evolution of bridge decks
- Flutter
- Quasi-static analysis
- Unsteady-aerodynamics

Map of the course

Wind Engineering

Fluid Dynamics

Numerical methods



At the end of the 19th century the evolution of material and construction science led to the construction of bridges with main span in the order of 500 m.



Brooklyn Bridge, main span 486 m, **has been the longest in the world from 1883-1903**. To be noticed is the hybrid suspension system (suspended and cable-stayed).

These bridges are designed with closed framed decks, which show high torsional stiffness (closed sections).



These torsionally-rigid decks are denoted as first generation decks.

Historical evolution: Golden Gate I CWE

Thanks to the improvement in material and construction technology, the main span kept increasing, despite some vibration problem. A notable example is the **Golden Gate, main span 1282 m, world longest between 1937-1964** (no hybrid suspension system here...).

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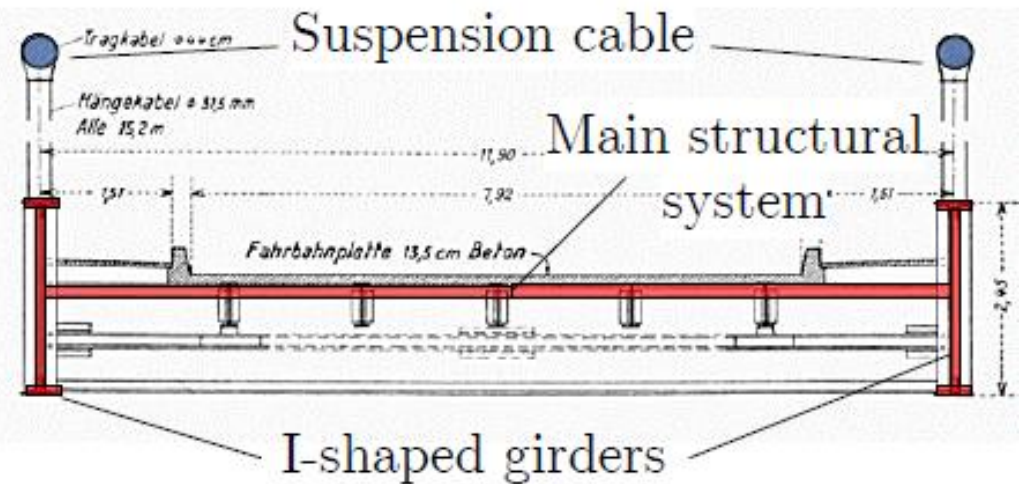


Historical evolution: Golden Gate II CWE

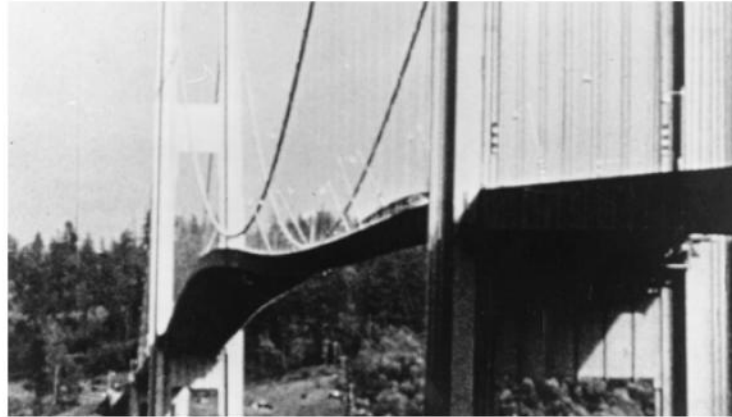
However, design was still based on the concepts used in the first generation of decks, **characterized by high stiffness.**



A few months after the opening of the Golden Gate, the First Tacoma Narrow Bridge was opened. The main span was (only) 853 m, but it was equipped with a **revolutionary light deck composed of an open section**.



The bridge was designed to withstand wind up to 33 m/s. During construction the bridge was nicknamed Galloping Gertie. It collapsed on 7 November 1940, 4 months after completion. Wind was 18 m/s.



The collapse happened after strong vibrations in torsion, with amplitudes up to approximately 35 degrees.

After the collapse of the First Tacoma Narrow bridge a **return to the torsionally stiff deck was inevitable**. An example is the Verrazzano bridge, main span 1298 m, longest in the world between 1964-1981.



Substantially, the first deck generation did not undergo major revisions for more than 50 years.

However, already in 1966 the Severn Bridge was completed in UK. The main span was 988 m, but it **was equipped with a revolutionary deck**.



The underlying idea is to control the production of aerodynamic forces by a careful aerodynamic design, rather than withstand them.

The deck is said to be “aerodynamic” or “aerodynamically transparent”. It is the first deck belonging of the second generation.

Historical evolution: Akashi Kaikyo CWE

Despite the success of the Severn Bridge, in 1998, when Japan built the Akashi Kaikyo bridge, 1991 m main span, **a cautions approach was taken, returning to the first generation design.**



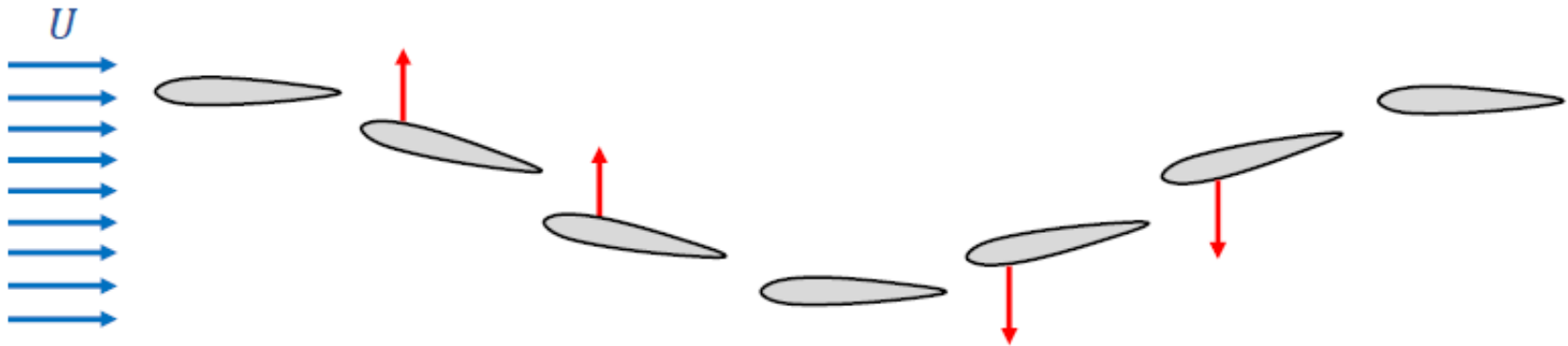
The Akashi bridge has been the longest in the world from 1998 to 2022.

In 2022 the Canakkale bridge was built in Turkey, with a main span of 2023 m. It is equipped with an innovative twin box deck section, initially developed for the Messina Bridge and already used for the Xihoumen Bridge in China in 2009.



This kind of deck strongly enhances aerodynamic performance, leading to a great reduction of aerodynamic forces and increased aerodynamic stability.

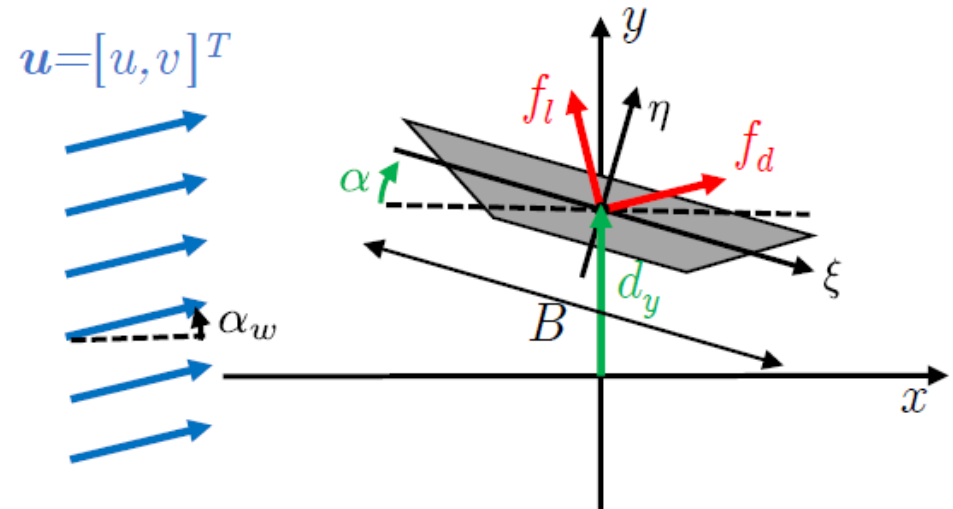
Actually the Tacoma Bridge is a quite particular case, and we do not detail it here for the sake of brevity. Modern decks are more likely prone to **classical flutter**, which is a more complex nuance of the aeroelastic instabilities which we already considered.



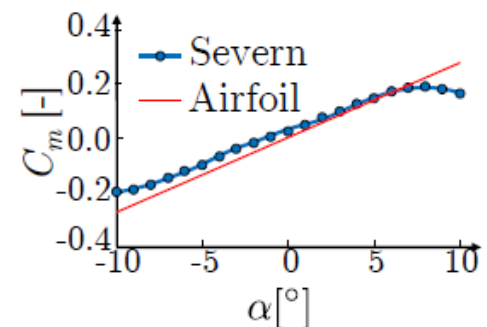
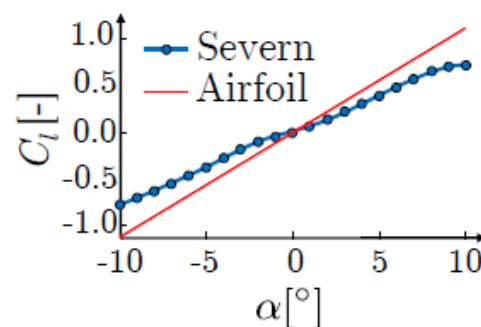
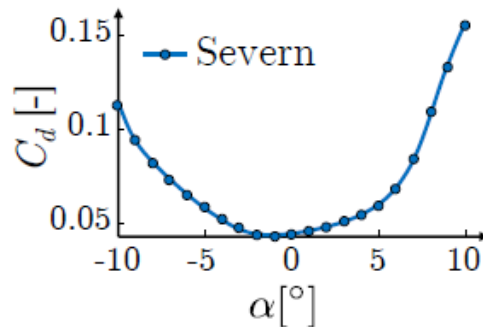
We now proceed to the simplest approach able to predict classical flutter. Such an approach has strong limitations, but it allows us to gain insight in the phenomenon.

We consider the bridge deck as a two degrees of freedom system.

and assume to know the aerodynamic coefficients



$$C_d = \frac{F_d}{1/2\rho U^2 B}, \quad C_l = \frac{F_l}{1/2\rho U^2 B}, \quad C_m = \frac{M}{1/2\rho U^2 B^2}.$$



The equations of motion can be written as

$$\begin{cases} m_{s,y}\ddot{d}_y + c_{s,y}\dot{d}_y + k_{s,y}d_y = f_y(t) \\ m_{s,\alpha}\ddot{\alpha} + c_{s,\alpha}\dot{\alpha} + k_{s,\alpha}\alpha = m(t) \end{cases}$$

which, in matrix notation read

$$\begin{bmatrix} m_{s,y} & 0 \\ 0 & m_{s,\alpha} \end{bmatrix} \begin{bmatrix} \ddot{d}_y \\ \ddot{\alpha} \end{bmatrix} + \begin{bmatrix} c_{s,y} & 0 \\ 0 & c_{s,\alpha} \end{bmatrix} \begin{bmatrix} \dot{d}_y \\ \dot{\alpha} \end{bmatrix} + \begin{bmatrix} k_{s,y} & 0 \\ 0 & k_{s,\alpha} \end{bmatrix} \begin{bmatrix} d_y \\ \alpha \end{bmatrix} = \begin{bmatrix} f_y(t) \\ m(t) \end{bmatrix}$$

or, synthetically

$$\underline{\mathcal{M}}_s \ddot{\underline{u}} + \underline{\mathcal{C}}_s \dot{\underline{u}} + \underline{\mathcal{K}}_s \underline{u} = \underline{\mathcal{F}}(t),$$

The angle of attack of the wind (relative velocity between fluid and deck) is

$$\alpha_{wr} = \text{atan} \left(\frac{v - \dot{d}_y}{u} \right)$$

so that the **total angle of attack** is

$$\alpha_e = \alpha + \text{atan} \left(\frac{v - \dot{d}_y}{u} \right),$$

It is then possible to calculate aerodynamic forces as

$$\mathcal{F}(t) = \begin{bmatrix} f_y(t) \\ m(t) \end{bmatrix} = q_r \begin{bmatrix} B [C_l(\alpha_e) \cos(\alpha) + C_d(\alpha_e) \sin(\alpha)] \\ B^2 C_m(\alpha_e) \end{bmatrix}, \quad q_r = \frac{1}{2} \rho |\mathbf{u}_r|^2,$$

As usual, we assume

$$\begin{bmatrix} u(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} U + u'(t) \\ v'(t) \end{bmatrix}$$

so that

$$q_r \approx \frac{1}{2} \rho |\mathbf{u}_r|^2 \approx U^2 + 2Uu' = \frac{1}{2} \rho U^2 \left(1 + 2 \frac{u'}{U} \right),$$

while linearising the angle of attack

$$\alpha_e \approx \alpha + \frac{v'}{U} - \frac{\dot{d}_y}{U}.$$

We can now proceed to expand the aerodynamic coefficients

$$C_d(\alpha_e) = C_d(0) + C'_d \alpha_e, \quad C_l(\alpha_e) = C_l(0) + C'_l \alpha_e, \quad C_m(\alpha_e) = C_m(0) + C'_m \alpha_e,$$

So that we finally have

$$\mathcal{F}(t) = \begin{bmatrix} f_y(t) \\ m(t) \end{bmatrix} = q \left(1 + 2\frac{u'}{U} \right) \begin{bmatrix} B [C_l(0) + C'_l \alpha_e + C_d(0) \alpha] \\ B^2 [C_m(0) + C'_m \alpha_e] \end{bmatrix},$$

which after reorganization and simplification of small terms becomes

$$\mathcal{F}(t) = \begin{bmatrix} f_y(t) \\ m(t) \end{bmatrix} = q \begin{bmatrix} B [C_l(0) + 2C_l(0)\frac{u'}{U} + C'_l \alpha_e + C_d(0) \alpha] \\ B^2 [C_m(0) + 2C_m(0)\frac{u'}{U} + C'_m \alpha_e] \end{bmatrix}.$$

which takes the form

$$\mathcal{F}(t) = \mathcal{F}_s + \mathcal{F}_b + \underline{\mathcal{C}}_{ae} \dot{\mathcal{U}} + \underline{\mathcal{K}}_{ae} \mathcal{U},$$

$$\mathcal{F}_{ta} = q \begin{bmatrix} BC_l(0) \\ B^2 C_m(0) \end{bmatrix}, \quad \mathcal{F}_b = q \begin{bmatrix} 2BC_l(0)\frac{u'}{U} + BC'_l \frac{v'}{U} \\ 2B^2 C_m(0)\frac{u'}{U} + B^2 C'_m \frac{v'}{U} \end{bmatrix},$$

$$\underline{\mathcal{C}}_{ae} = q \begin{bmatrix} -\frac{C'_l B}{U} & 0 \\ -\frac{C'_m B^2}{U} & 0 \end{bmatrix}, \quad \underline{\mathcal{K}}_{ae} = q \begin{bmatrix} 0 & (C'_l + C_d(0))B \\ 0 & C'_m B^2 \end{bmatrix},$$

Our goal is now to study the characteristics (mainly the damping) of such two degrees of freedom system.

We make a step back and consider a single dof system

$$m_u \ddot{\mathcal{U}} + c_u \dot{\mathcal{U}} + k_u \mathcal{U} = f_u(t).$$

This is a second order system, which can be reduced to a first order system introducing $\mathcal{V} = \frac{d\mathcal{U}}{dt}$

$$\begin{bmatrix} 1 & 0 \\ 0 & m_u \end{bmatrix} \begin{bmatrix} \dot{\mathcal{U}} \\ \dot{\mathcal{V}} \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ k_u & c_u \end{bmatrix} \begin{bmatrix} \mathcal{U} \\ \mathcal{V} \end{bmatrix} = \begin{bmatrix} 0 \\ f_u \end{bmatrix}$$

which can be written as

$$\underline{\mathcal{A}}\dot{\underline{\mathcal{W}}} + \underline{\mathcal{B}}\underline{\mathcal{W}} = \underline{\mathcal{F}}(t),$$

The equation can be easily extended to multiple degrees of freedom

$$\mathcal{A}\dot{\mathcal{W}} + \mathcal{B}\mathcal{W} = \mathcal{F}(t),$$

$$\mathcal{W} = \begin{bmatrix} \mathcal{U} \\ \mathcal{V} \end{bmatrix}, \quad \mathcal{A} = \begin{bmatrix} I & 0 \\ 0 & \mathcal{M} \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0 & -I \\ \mathcal{K} & \mathcal{C} \end{bmatrix}, \quad \mathcal{F}(t) = \begin{bmatrix} 0 \\ f(t) \end{bmatrix},$$

We assume the solution to be of the type

$$\mathcal{W} = \tilde{\mathcal{W}}e^{\lambda t}.$$

Substituting into the governing equation

$$[\lambda\mathcal{A} + \mathcal{B}]\tilde{\mathcal{W}}e^{\lambda t} = 0.$$

This system of equations shall be respected for every t , so

$$\mathcal{B}\tilde{\mathcal{W}} = -\lambda\mathcal{A}\tilde{\mathcal{W}}.$$

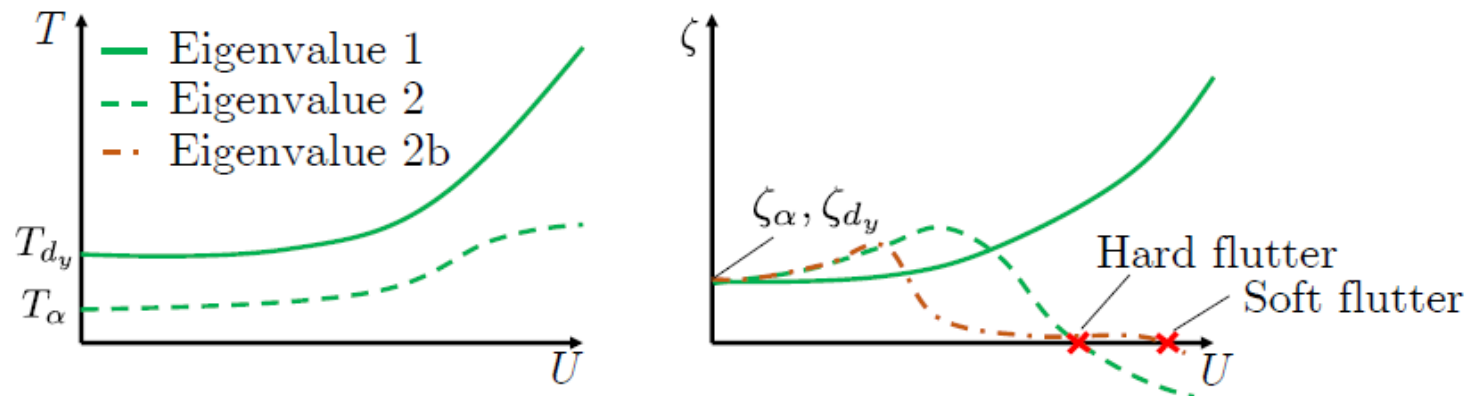
This is a **generalized eigenvalue problem**, which can be solved providing the system eigenvalues λ_j (and eigenvectors). Going back to the assumed solution

$$\mathbf{w} = \tilde{\mathbf{w}} e^{\lambda t}.$$

We have that

$$T_j = \frac{2\pi}{\Im(\lambda_j)}, \quad \zeta_j = -\frac{\Re(\lambda_j)}{|\lambda_j|},$$

Tracking these eigenvalue at different wind speed, we obtain



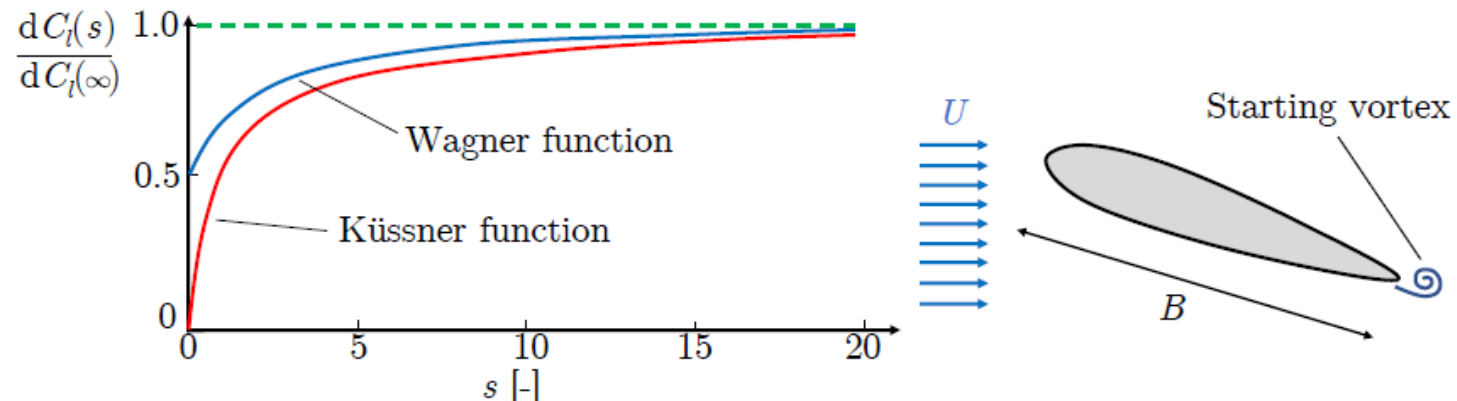
The proposed approach has numerous shortcomings

- 1) misses **inertial effects**.
- 2) misses some **damping effects** (in still air);
- 3) it is actually known from airfoil theory that the **effective angle of attack incorporates the torsional velocity**

$$\alpha_e = \alpha + \boxed{\frac{1}{4} \frac{B}{U} \dot{\alpha}} + \frac{v'}{U} - \frac{\dot{d}_y}{U}.$$

but no general formulation for deck exists.

- 4) **memory effects**



More in general, aeroelastic forces on **vibrating decks** can be approximated as

$$\mathcal{F}_{ae} = \frac{1}{2}\rho U^2 \begin{bmatrix} B \left(K^2 H_4^* \frac{d_y}{B} + K H_1^* \frac{\dot{d}_y}{U} + K^2 H_3^* \alpha + K H_2^* \frac{\dot{\alpha} B}{U} \right) \\ B^2 \left(K^2 A_4^* \frac{d_y}{B} + K A_1^* \frac{\dot{d}_y}{U} + K^2 A_3^* \alpha + K A_2^* \frac{\dot{\alpha} B}{U} \right) \end{bmatrix}$$

$$U_{red} = \frac{U}{Bf} \qquad K = \frac{2\pi}{U_{red}}$$

where

$$H_i^* = H_i^*(U_{red}) \qquad A_i^* = A_i^*(U_{red})$$

are known as **flutter derivatives**, which must be obtained from wind tunnel tests or numerical simulations. Their evaluation, requires to consider moving bridge decks. Their use makes the eigenvalue analysis previously described nonlinear.